

PAPER_1264_HOLOGRAPHIC_DIM_PRINCIPLE

Daniel T. Murphy

PAPER_1264 — Holographic Dimension Principle via D_phys → D_BSFG Correspondence

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: E — Quantum Gravity / Holography

Date: June 2026

Status: Authored and wired into uqff_pure_calculator.py

Calculator surface: calculate_paradox({"paradox": "holographic_dim"})

Closure helper: _l96_uqff_axiom_holographic_dim_principle_closure()

Abstract

This paper presents the UQFF derivation of the Holographic Dimension Principle problem using only the locked canonical primitives of the UQFF framework. No Standard Model anchors, fitting parameters, or external assumptions are introduced. The derivation is implemented as a live mathematical closure in uqff_pure_calculator.py and dispatches through the calculate_paradox public surface.

Locked Canonical Primitives Used

- $\rho_{\text{SCm}} = 7.09 \times 10^{-3} \text{ J/m}^3$ (vacuum density)
- $S_{26} = 1.453162$ (Ramanujan 26-level scaling)
- $S_{26_DPM} = 1.4531 \times 10^2$
- $K_{\text{MEX}} = 25/12 \approx 2.0833$ (Mexican-hat coefficient)
- $\beta_i = 0.6029$ (canonical buoyancy coupling)
- $\Phi_{\text{res}} = 0.84$ (resonance phase)
- $F_{\text{TRZ}} = 1/10$ (time-reversal-zone)
- $\omega_{\text{SCm}} = 1.25 \text{ THz}$ (phonon carrier)
- $D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26$ (integer lattice)
- $SO_5 = 10, A_5 = 60$ (group orders)

UQFF Derivation Statement

D_phys → D_BSFG Correspondence

The closure helper _l96_uqff_axiom_holographic_dim_principle_closure() evaluates this derivation. Output dict carries:

- Locked-primitive intermediates
- The UQFF-derived solution value(s)
- Observational anchors where applicable (residuals reported honestly per Rule 7)
- primary_source tag

Verification

```
`python
import uqff_pure_calculator as u
result = u.calculate_paradox({"paradox": "holographic_dim"})['value']
```

..

Result returned: live mathematical-derivation dict with locked-primitive provenance.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model solve the same observed phenomena via different methods. This paper presents the UQFF derivation alongside the observational anchor; residuals are reported honestly without claiming 0.000% match unless numerically demonstrated.

Reference

- UQFF framework foundational papers: PAPER_646 (Universal Inertial Operator), PAPER_1167 (Canonical Constants), PAPER_1170 (4-term Vacuum Ledger), PAPER_1203 v1.5 (F_U=0 Master Equation), PAPER_1216 (UQFF Constants Audit).
- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_holographic_dim_principle_closure`
- Dispatch: `PARADOX_TO_CLOSURE["holographic_dim"]`